

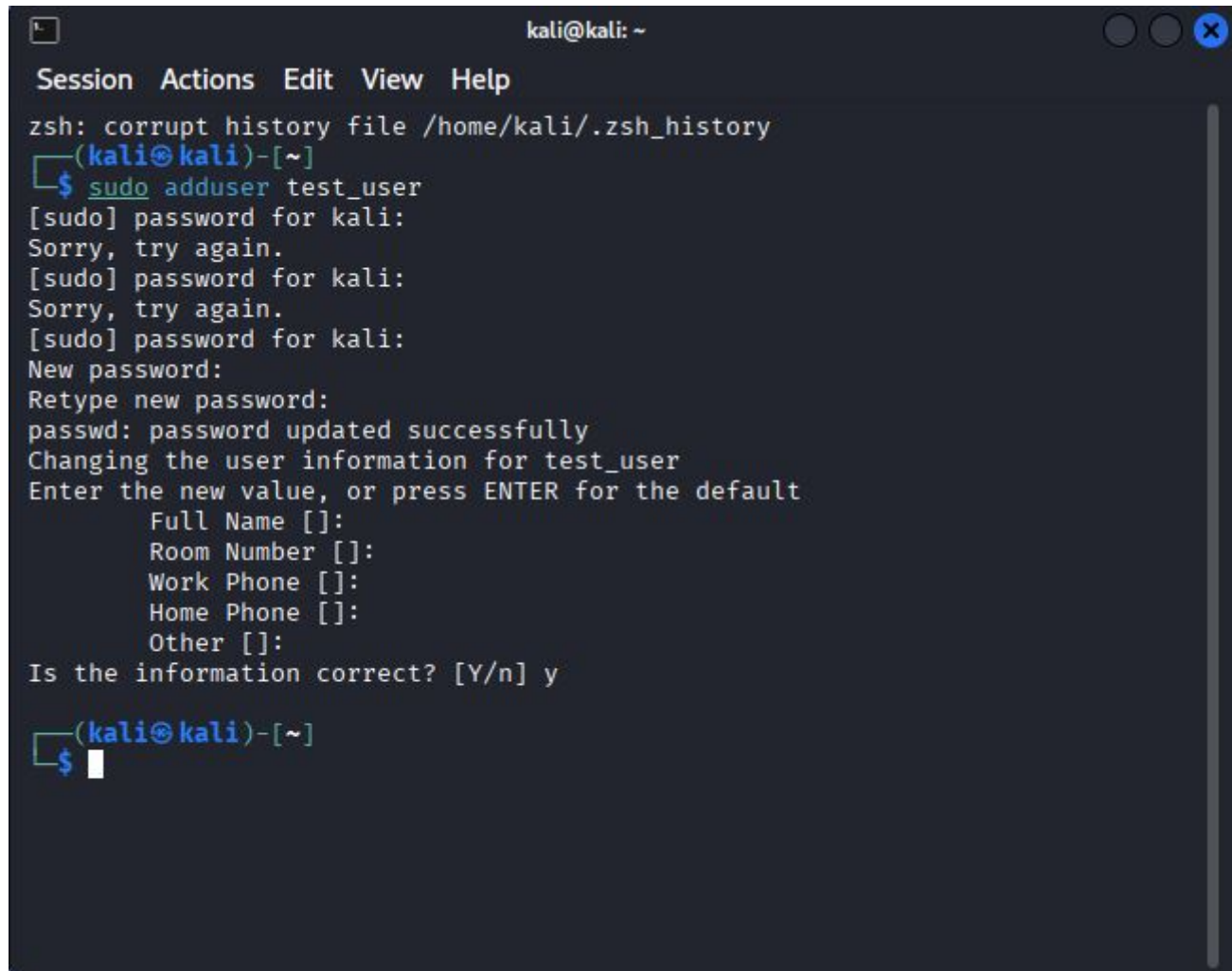
ESERCIZIO: Authentication cracking con Hydra

In questo esercizio nel laboratorio verrà utilizzata la VM kali in modalità sia da attaccante che da bersaglio.

Per poterlo eseguire viene creato l'utente che dovrà essere craccato.

Utente: test_user

password: testpass

A terminal window titled 'kali@kali: ~' with a menu bar 'Session Actions Edit View Help'. The terminal shows a message 'zsh: corrupt history file /home/kali/.zsh_history' followed by a prompt '(kali@kali)-[~]'. The user enters '\$ sudo adduser test_user'. The system prompts for the password for kali, which is entered three times with 'Sorry, try again.' messages. Then it asks for a new password and to retype it. A message 'passwd: password updated successfully' is shown. Next, it says 'Changing the user information for test_user' and 'Enter the new value, or press ENTER for the default'. It then prompts for Full Name, Room Number, Work Phone, Home Phone, and Other, all of which are left empty. Finally, it asks 'Is the information correct? [Y/n] y'. The terminal ends with the prompt '(kali@kali)-[~]' and '\$' with a cursor.

```
kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ sudo adduser test_user
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for test_user
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
(kali@kali)-[~]
$
```

Poi ho attivato il servizio SSH e verificato che fosse funzionante

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ sudo service ssh start  
  
(kali@kali)-[~]  
$ sudo service ssh status  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: >  
   Active: active (running) since Fri 2026-01-16 05:21:08 EST; 40s ago  
 Invocation: ea194000c01a4aa5a24aee5089b20bc1  
    Docs: man:sshd(8)  
          man:sshd_config(5)  
 Process: 55308 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUC>  
Main PID: 55310 (sshd)  
   Tasks: 1 (limit: 9179)  
  Memory: 2M (peak: 2.6M)  
    CPU: 59ms  
   CGroup: /system.slice/ssh.service  
           └─55310 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup>  
  
Jan 16 05:21:08 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell>  
Jan 16 05:21:08 kali sshd[55310]: Server listening on 0.0.0.0 port 22.  
Jan 16 05:21:08 kali sshd[55310]: Server listening on :: port 22.  
Jan 16 05:21:08 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell >
```

Ho Testato la connessione in SSH

```
test_user@kali: ~  
Session Actions Edit View Help  
permitted by applicable law.  
(test_user@kali)-[~]  
$ exit  
logout  
Connection to localhost closed.  
  
(kali@kali)-[~]  
$ ssh test_user@10.0.2.15  
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.  
ED25519 key fingerprint is: SHA256:HQsB5hYbqIrf1DZf7Tyv7NvD5Sh6ly4FLJatg+NDft  
c  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:4: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.2.15' (ED25519) to the list of known hosts.  
test_user@10.0.2.15's password:  
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-  
09-24) x86_64  
  
The programs included with the Kali GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Fri Jan 16 05:29:23 2026 from ::1  
(test_user@kali)-[~]  
$ █
```

CONFIGURAZIONE HYDRA

In questa fase ho utilizzato le opzioni (-l -p in minuscolo) per testare una singola coppia di credenziali con un comando mirato: `hydra -l test_user -p testpass 10.0.2.15 -t 2 ssh`

```
(kali@kali)-[~]  
$ hydra -l test_user -p testpass 10.0.2.15 -t 2 ssh  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in  
military or secret service organizations, or for illegal purposes (this is n  
on-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-16 06:  
24:56  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try  
per task  
[DATA] attacking ssh://10.0.2.15:22/  
[22][ssh] host: 10.0.2.15 login: test_user password: testpass  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-01-16 06:  
24:57  
  
(kali@kali)-[~]  
$ █
```


Nell'espletare tutto l'esercizio ho poi scaricato una lista di username e password (comando: `sudo apt install seclists`) per poter procedere alla **SECONDA FASE**. Come indicato ho effettuato una configurazione di un servizio FTP tramite vsftpd in modo da esporre un servizio di rete con autenticazione e successivo TEST della robustezza delle credenziali tramite HYDRA.

```
kali@kali: ~  
Session Actions Edit View Help  
  
Installing:  
vsftpd  
  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 981  
  Download size: 145 kB  
  Space needed: 356 kB / 41.2 GB available  
  
Get:1 http://kali.download/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.4 [145 kB]  
Fetched 145 kB in 0s (328 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package vsftpd.  
(Reading database ... 460512 files and directories currently installed.)  
Preparing to unpack .../vsftpd_3.0.5-0.4_amd64.deb ...  
Unpacking vsftpd (3.0.5-0.4) ...  
Setting up vsftpd (3.0.5-0.4) ...  
/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy director  
y /var/run/, updating /var/run/vsftpd/empty → /run/vsftpd/empty; please updat  
e the tmpfiles.d/ drop-in file accordingly.  
update-rc.d: We have no instructions for the vsftpd init script.  
update-rc.d: It looks like a network service, we disable it.  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for kali-menu (2025.4.2) ...  
  
(kali@kali)-[~]  
$
```

Verificato poi che l'FTP fosse attivo

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ftp 10.0.2.15  
Connected to 10.0.2.15.  
220 (vsFTPd 3.0.5)  
Name (10.0.2.15:kali): test_user  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> exit  
221 Goodbye.  
  
(kali@kali)-[~]  
$
```

In questo caso ho specificato ad Hydra di utilizzare il modulo FTP per risalire sempre alle credenziali

```
kali@kali: ~  
Session Actions Edit View Help  
220 (vsFTPd 3.0.5)  
Name (10.0.2.15:kali): test_user  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> exit  
221 Goodbye.  
  
(kali@kali)-[~]  
$ hydra -l test_user -p testpass 10.0.2.15 ftp  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in  
military or secret service organizations, or for illegal purposes (this is n  
on-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-16 08:  
09:23  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try  
per task  
[DATA] attacking ftp://10.0.2.15:21/  
[21][ftp] host: 10.0.2.15 login: test_user password: testpass  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-01-16 08:  
09:24  
  
(kali@kali)-[~]  
$
```

CONCLUSIONE: L'esercizio ha permesso di verificare il funzionamento di Hydra su due protocolli diversi SSH (Prendere il controllo del terminale) e FTP (Gestire, rubare o caricare file) evidenziando come lo stesso metodo di attacco possa essere utilizzato su 2 meccanismi di sicurezza differenti.